

AeroGrid Engineering Solutions: Telemetry infrastructure evaluation

Data processing findings:

After running the decoupled streaming data pipeline against the active wind turbine dataset, individual assets that exceeded safe operating criteria were effectively separated. Turbines that required immediate engineering maintenance intervention were identified using two key metrics: establishing an average temperature run-rate over 85.0°C or identifying a single peak transient vibration spike greater than 15.0 mm/s.

Architectural justification:

AeroGrid's old architecture has major ingestion degradation during peak weather events due to the strong monolithic coupling between streaming network connections and the file-writing database layer. When heavy winds blow and hundreds of physical sensors overload the system, the server loses packets and fails.

To overcome this single-server bottleneck, the re-engineered design implements a distributed message queue that uses Apache Kafka (or AWS Kinesis) as an enterprise "shock absorber." When high-velocity anomalies overwhelm the network, Kafka safely registers and queues incoming raw CSV/JSON data packets within a fault-tolerant cluster.

The stateless, serverless computing blocks (AWS Lambda) then ingest records asynchronously from the queue in micro-batches to assess safety rules on the fly. This entirely separates data gathering and processing, ensuring 100% ingestion uptime despite variable network surges.

Cost optimization strategy:

Retaining enormous historical high-resolution operational streams in current transactional databases incurs unsustainable cloud infrastructure expenses. The suggested solution addresses this issue by utilising a contemporary tiered storage framework:

Hot Storage Tier (Time-Series Database): Stores just the most recent 30 days of telemetry data to efficiently handle active read/write operational monitoring and dashboards (e.g., InfluxDB or AWS Timestream).

Cold Storage Tier (Object Storage Data Lake): Automated object lifecycle management procedures regularly compress and transfer data older than 30 days into extremely low-cost object storage (AWS S3 or Azure Blob Storage). This retains deep history logs for long-term predictive machine learning models at a cost that is more than 90% lower than retaining them in an active database.